

Programme	BSc Computer Science				
Course Title	Cyber Forensic				
Type of Course	Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Understanding concept Computer Hardware, Operating System 2. Knowledge of information security concepts, including confidentiality, integrity, and availability 3. Knowledge of legal and ethical issues surrounding digital evidence collection, preservation, and analysis is crucial for conducting forensic investigations in compliance with applicable laws and regulations.				
Course Summary	This course provides an overview of cyber forensics and cyber laws, focusing on the principles, techniques, and legal considerations involved in investigating cybercrimes, preserving digital evidence, and navigating legal frameworks governing cybersecurity.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand the fundamental concepts, principles, and methodologies of cyber forensics	Ap	C	
CO2	To develop skills in acquiring preserving, and analysing digital evidence from various source	Ap	P	
CO3	To learn and understand techniques and tools to investigate cybercrimes, security incidents, and data breaches.	Ap	P	
CO4	Demonstrate proficiency in conducting network, disk, memory, and mobile device forensics examinations.	Ap	P	
CO5	Evaluate ethical, legal, and privacy considerations in cyber forensics investigations and evidence handling.	E	M	
CO6	Apply critical thinking, problem-solving, and decision-making skills to address challenges in cyber forensics and cybersecurity.	Ap	P	
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus

Module	Unit	Content	Hrs (48+12)	Marks
I	INTRODUCTION TO CYBER FORENSICS		10	15
	1	Computer Forensics Fundamentals: What is Computer Forensics? Use of Computer Forensics in Law Enforcement, Computer Forensics Services	2	
	2	Computer Forensics Assistance: Human Recourses/Employment Proceedings, Benefits of professional Forensics Methodology, Steps taken by Computer Forensics Specialists	2	
	3	Computer Forensics Technology: Business Computer Forensic Technology, Military Computer Forensic Technology, Law Enforcement	2	
	4	Vendor and Computer Forensics Services: Types of services provided by vendors, Criteria for selecting a computer forensics vendor, Vendor Engagement and Contracts, Evaluation of vendor capabilities, expertise and reputation	2	
	5	Cyber forensics tools and case studies: Disk Imaging (EnCase, FTK), File Analysis (FileInsight and ExifTool),	2	
II	COMPUTER FORENSICS EVIDENCE		10	15
	6	Computer forensics evidence and capture: Why Collect Evidence, Types of Evidence, The Rules of Evidence, Volatile Evidence,	2	
	7	Data Recovery: Definition, Data Back-up and Recovery, The Role of Back-up in Data Recovery, The Data -Recovery Solution	2	
	8	General Procedure for Data Collection: Collection and Archiving, Methods of Collection	2	
	9	Identification of Data: Timekeeping, Forensic Identification and Analysis of Technical Surveillance Devices, Reconstructing Past Events.	2	
	10	Controlling Contamination: The Chain of Custody, Reconstructing the Attack, The digital crime scene, Investigating Cybercrime, Investigating Web attacks, Investigating network Traffic	2	
III	FORENSIC ANALYSIS AND VALIDATION		14	20
	11	Computer image Verification and Authentication: Special needs of Evidential Authentication,	2	
	12	Computer forensic analysis: Determining what data to collect and analyse, validating forensic data, addressing data-hiding techniques, performing remote acquisitions	2	
	13	Computer forensic validation: Validating forensic data, addressing data-hiding techniques, performing remote acquisitions	2	
	12	Network Forensics: Network forensic overview, Performing live acquisitions, Developing standard procedures for network forensics	2	
	13	Network Forensic Tools: Overview, Wireshark, tcpdump, and NetworkMiner, Network Traffic Analysis Tools	2	

	14	Ethical Hacking: Essential Terminology, Windows Hacking, Malware, Scanning, Cracking.	2	
	15	Tactics of the Military, Tactics of Terrorist and Rogues, Tactics of Private Companies	2	
IV	CYBER CRIME AND CYBER LAW		14	
	16	Mobile device forensics: Understanding mobile device forensic, understanding acquisition procedures for cell phones and mobile devices.	2	20
	17	Cyber Crimes: Types of cybercrimes against individuals and institution, States-various offenses and punishments	2	
	18	Digital Signature: Concepts of public key and private key, Certification Authorities and their role, Creation and authentication of digital signature.	2	
	19	E-contracting: Features of E-contracts, Formation of E-contracts and types	2	
	20	E-governance: E-governance models, E-commerce- salient features and advantages.	2	
	21	Cyber Law: Understanding cyber space, Defining cyber law, Scope and jurisprudence	2	
	22	Indian Cyber Law: Overview of Indian legal system, Introduction to IT Act 2000, Amendment in IT Act.	2	
V	Open Ended Module- Trends in Software Engineering			
		1. Case Study. 2. Simulate real-world cyber incidents and develop incident response plans. 3. An activity that emphasizes teamwork, communication, and decision-making under pressure. 4. Work on a comprehensive cyber forensics project that integrates concepts from multiple areas of study. 5. Apply forensic techniques to investigate a real or simulated cyber incident and produce a detailed report.		

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	-	-	-	2	1						
CO 2	2	-	-	-	2	1						
CO 3	2	-	-	-	3	2						
CO 4	1	-	-	-	1	1						
CO 5	2	--	-	--	3	1						
CO 6	3		-		2	1						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓

References:

1. Computer Forensics: Computer Crime Scene Investigation, 2nd Edition, John R. Vacca, Charles River Media, 2005
2. Cyber Forensics - Concepts and Approaches, Ravi Kumar & B Jain, 2006, ICFAI university press
3. Understanding Cryptography: A Textbook for Students and Practitioners, Christof Paar, Jan Pelzl, 2010, Second Edition, Springer's.
4. Live Hacking: The Ultimate Guide to Hacking Techniques & Countermeasures for Ethical Hackers & IT Security Experts, Ali Jahangiri, First edition, 2009
5. Computer Forensics: Investigating Network Intrusions and Cyber Crime (Ec-Council Press Series: Computer Forensics), 2010